SESSION 13 – DECISION TREE (REGRESSION)

🎯 **Topic**

- Decision Tree for regression
- Difference: classification vs regression tree
- Splitting based on variance (intuition)
- Piecewise prediction
- Overfitting in regression trees
- Practical implementation

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand how Decision Trees work for **regression problems**
- Differentiate **classification tree vs regression tree**
- Learn how trees split data using **variance reduction (intuition)**
- Interpret **step-wise (piecewise) predictions**
- Build a **Decision Tree Regressor**
- Understand **overfitting control**

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"In the last session, we used Decision Tree for classification (Pass/Fail).
Today we will use the same idea to predict numbers like salary or price."

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3. 📖 What is Decision Tree (Regression)?

3.1 Definition

A Decision Tree Regressor predicts **continuous values** by splitting data into regions and assigning an average value to each region.

3.2 Simple Meaning (Most Important Line)

"Decision Tree Regression splits data and predicts a constant value for each region."

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4. 🔍 Classification vs Regression Tree

Feature	Classification Tree	Regression Tree
Output	Category (0/1)	Number

Feature	Classification Tree	Regression Tree
Example	Pass/Fail	Salary
Goal	Classify	Predict value

Teaching Line

"Same tree structure, different output type."

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5. 📊 How Decision Tree Regression Works

Step-by-Step

1. Choose best feature to split
2. Split data into regions
3. Continue splitting
4. At leaf node:
   👉 Predict **average value of that region**

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6. 📊 Splitting Logic (Intuition Only)

Concept: Variance Reduction

👉 Goal:

- Reduce variation in each split

Meaning

- Data inside each region should be **similar**

Teaching Line

"Tree splits data so that values in each group are as similar as possible."

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7. 📊 Example (Very Important)

Experience vs Salary

If Experience < 5 → Salary ≈ 35,000
If Experience ≥ 5 → Salary ≈ 90,000

👉 Instead of smooth curve:

- Tree gives **step-like prediction**

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8. 📈 Output Shape (Important Insight)

Linear/Polynomial Regression:

- Smooth curve

Decision Tree:

- Step function

Teaching Line

"Decision Tree predicts in steps, not smooth curves."

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9. ⚠️ Overfitting in Regression Trees

Problem

- Too many splits
- Very specific regions

👉 Model memorizes data

Solution

- Limit max depth
- Minimum samples per split
- Pruning

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10. 📊 Advantages

✅ Handles non-linear data
✅ No scaling required
✅ Easy to understand
✅ Works well on complex patterns

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11. ⚠️ Limitations

❌ Overfitting risk
❌ Not smooth predictions
❌ Sensitive to small data changes

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12. 🎯 Interview Concepts

Q1: How does Decision Tree Regression work?

👉 Splits data and predicts average value.

Q2: What is variance reduction?

👉 Reducing variability in splits.

Q3: Why output is step-like?

👉 Because each region has constant value.

Q4: How to control overfitting?

👉 Limit depth, pruning.

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13. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Predict Salary using Decision Tree Regression

14. 📊 Output Interpretation

Explain:

- Predictions are not smooth
- Tree creates ranges
- Each range has constant value

👉 Example:

- 1–4 years → similar salary
- 5–8 years → higher salary

👉 Conclusion:

“Decision Tree approximates data using step-wise predictions.”

Decision Tree Regression is powerful for capturing non-linear patterns, but it predicts values in steps. Proper control of tree depth is important to avoid overfitting.